

WRITTEN WORK GUIDELINES AND EXPECTATIONS

Linear Algebra, Spring 2019

WRITTEN WORK

A goal of this course is to improve students' ability to interpret, organize, and to communicate quantitative ideas. To encourage this improvement, you will be asked to write solutions to written homework problems, verify the correctness of these solutions using a key, and create a professional, readable, fluent, and valid final report of your corrected findings. Ideally, this document will act as a rough set of guidelines and as an informative tool on how to write professional quality mathematical documents.

READABILITY

Your written work will be examined for readability. When determining the readability of your report consider asking yourself these questions: Does it look like something you would read? Is it typed? Is the same spacing, indent, font, and font size used consistently throughout the report? Did you use a single voice throughout the report? Is that voice your own?

FLUENCY

Part of every assessment will focus on your ability to incorporate your mathematical thoughts using algebraic representation (functions and equations), numerical representations (tables and numbers), graphical displays (pictures), and correct established notation and terminology into your written work. Strings of equations with no explanation or motivation do not constitute a fluent response and, although a carefully chosen, motivated, and explained figure may be worth a thousand words, a similar number of figures—purely in and of themselves—are worth nothing without the narrative scaffolding of a proper explanatory context. When determining the fluency of your report, consider asking yourself these questions: Does it look like a mathematical report? Are the functions and equations, tables and numbers, and pictures properly motivated and do they fit seamlessly into the structure of the report? Is the mathematical text produced using a mathematical text editor? Are the graphics of high quality and consistent?

VALIDITY

If an explanation requires calculations or entails a string of deductions, the extent to which your solution is valid will be analyzed. Validity corresponds to the discernible extent to which your method used is appropriate, your calculations are correct, and your deductions follow the rules of logic. When determining the validity of your report consider asking yourself these questions: Is your reasoning complete and correct for the problem? Did you choose a strategy that was logical? Did you fully explain the steps taken to produce your answer? Did you properly and rigorously define any terminology you used which deviates from the common, established notation used elsewhere in the course?